

Aditya Kulkarni

adityakulkarnius@gmail.com | (281) 876-7066 | linkedin.com/in/aditya-kulkarni-355b81217 | kulkarniaditya.com | github.com/Aditya-k24
| Raleigh, NC (Remote)

PROFESSIONAL SUMMARY

MS CS candidate (NC State, May 2026) with production Python and agentic AI engineering experience across LLM orchestration, event-driven pipelines, and full-stack TypeScript applications. At Isomer AI, built multi-provider LLM agent tooling across Anthropic Claude, OpenAI GPT-4o, and Google Gemini using Temporal.io workflow orchestration -- 394 commits, 10+ features, 7 production deployments. Designed event-driven AI platforms handling 2,000 async inference requests/second and ETL pipelines at Capgemini using Kafka + Spark. Proficient in Python, FastAPI patterns, TypeScript, Docker, PostgreSQL, and Kubernetes; experienced in CI/CD, distributed systems, and production observability.

CORE COMPETENCIES

Python / FastAPI

Agentic AI & LLM Orchestration

Event-Driven Pipelines

TypeScript / React

Docker / Kubernetes

PostgreSQL / MySQL / MongoDB

ETL Pipelines

CI/CD & Observability

WORK EXPERIENCE

Isomer AI May 2025 - Present

Product Engineer Intern — Chicago, IL (Remote)

- Implemented **agentic AI systems and LLM orchestration** across 3 providers and 8+ models (Anthropic Claude, OpenAI GPT-4o, Google Gemini) using MCP and Temporal.io workflow engine; built provider-agnostic abstraction with retry/fallback logic for production claims automation
- Shipped 10+ backend features in TypeScript/Node.js/PostgreSQL/Redis including RBAC, real-time event pipelines, and document processing (PDF, email via Microsoft Graph API, S3) -- 394 commits, 7 production deployments with Docker, GitHub Actions CI/CD, and Sentry observability

NC State University Apr 2025 - Present

Graduate Research Assistant — Raleigh, NC

- Designed a cloud-based genomic data platform on Google Cloud Run serving 500GB datasets to 100+ researchers; applied hyperparameter tuning and time-series analysis improving ML model accuracy by 12%

Mphasis (HP Company) Jun 2024 - Aug 2024

Software Developer Intern — Mumbai, India

- Built 5+ **Python AI workflows** including health claim automation (OCR, NLP, ML); improved accuracy by 25%; automated deployments on AWS and GCP with Grafana monitoring

Capgemini Jun 2023 - Aug 2023

AI/ML Intern — Mumbai, India

- Built distributed **ETL pipelines** using Hadoop + Spark for batch processing and Kafka for real-time ingestion; reduced ticket-processing latency by 20% and processing time by 15%

PROJECTS

LLM Eval Harness Python

Built a vendor-neutral benchmark in Python comparing Claude Sonnet 4.6, GPT-4o, and Gemini 2.5 Pro across 2,000-example frozen test set. Metrics: ExactMatch, TokenF1, grounded hallucination rate, p50/p95 latency, cost/call with paired bootstrap CIs and McNemar testing. Per-vendor adapter protocol isolates billing semantics and rate-limit signals.

Python, Anthropic/OpenAI/Google SDKs, DuckDB, Parquet, Streamlit, Docker

AlgoPulse – Agentic AI Orchestration Platform Event-Driven

Identified API-polling bottleneck under load; re-architected to event-driven model -- Supabase Edge Functions stream AI reasoning loops directly to clients. Temporal.io manages durable workflow state; Kafka handles worker coordination and fault-tolerant message brokering: 2,000 async inference requests/second without state loss.

TypeScript, Node.js, React, Supabase Edge Functions, Temporal.io, Kafka

PulseRank – Real-Time Leaderboard & Activity Feed

Multi-DB

Event-driven platform with MySQL for transactional writes (idempotency key, outbox pattern), MongoDB for feed document projections with TTL expiry, and Redis sorted sets for sub-millisecond leaderboard reads. Live updates via Server-Sent Events. Testcontainers integration tests, k6 benchmarks, OpenTelemetry + Prometheus observability, GitHub Actions CI/CD.

TypeScript, NestJS, Next.js, MySQL, MongoDB, Redis, Docker, OpenTelemetry, Prometheus

Glance AI – Privacy-First RAG System

Python + AI

Built a compliance-ready RAG pipeline for fintech/healthcare with per-tenant data isolation at the vector retrieval layer -- prevents cross-tenant context leakage. Deterministic output architecture ensures factual correctness for regulated data environments.

Python, LLMs, Vector Databases, RAG

EDUCATION

North Carolina State University — MS Computer Science

Aug 2024 - May 2026

Courses: Artificial Intelligence, Data Science, Algorithms, Operating Systems, Software Engineering, Object-Oriented Development

Dwarkadas J. Sanghvi College of Engineering — BTech Computer Science

Aug 2020 - May 2024

CGPA: 9.32/10 · Courses: Machine Learning, Big Data Infrastructure, DS & Algorithms, Data Warehousing, Software Engineering

SKILLS

Languages: Python, TypeScript, JavaScript, Java, SQL, C, R

Backend & AI: FastAPI, Node.js, NestJS, Temporal.io, Kafka, Spark, Hadoop, LangChain, MCP, OpenAI API, Anthropic Claude, Hugging Face, RAG

Frontend: React, Next.js, Vue (familiar), TypeScript, Tailwind CSS

Infra & Cloud: Docker, Kubernetes, Helm, Terraform, AWS, GCP (Cloud Run), GitHub Actions, OpenTelemetry, Prometheus

Databases: PostgreSQL, MySQL, MongoDB, Redis, Snowflake (familiar)